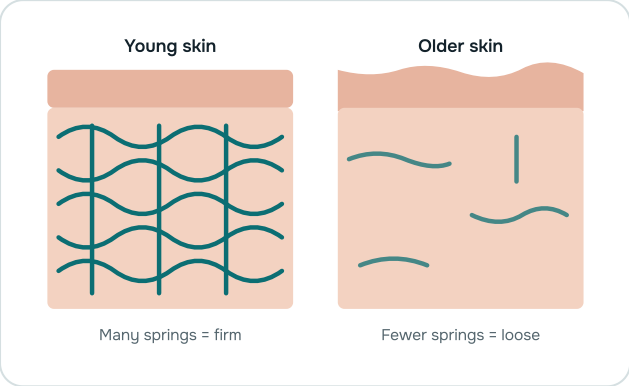
What is collagen, and why do people take it?

Before you compare products, it helps to know what collagen is, what it does in the body and what a collagen stick is.



1 **It is a protein**

Collagen is the most common protein in the human body.[1] It forms long, strong threads that hold skin, bones, joints, hair and nails together.

2 **It keeps skin firm**

In the skin, collagen threads work like the springs in a mattress. They keep skin firm, smooth and able to spring back.

3 **The body makes less with age**

As we get older, skin cells make less collagen than they did in youth.[2] Sun, smoking, stress and poor sleep add to this. Skin becomes thinner and drier, and lines appear.

4 **What a collagen stick is**

A stick is a small packet with one daily serving. You mix the powder with water, or eat the jelly straight from the stick. Inside is collagen that has been cut into small pieces, called collagen peptides.

5 **How it works**

The body absorbs some of the small pieces into the blood.[3] Lab studies suggest some of them may act as a signal that tells skin cells to make more of their own collagen.[4] In human studies, people took collagen every day, often for 8 to 12 weeks, before results were measured.[67]

There are different types of collagen. For skin, Type I matters most.

Type I

Skin, hair, nails and bones. About 90% of all collagen in the body. Comes from **fish** or cows. Fish collagen is mostly Type I and contains no pork or beef.

Type II

Joints and cartilage. Usually made from chicken. Used in joint products, not in skin products.

Type III

Skin and blood vessels. Works together with Type I. Usually made from cows or pigs.

WHAT IS "DA"

What does "Da" mean, and why is smaller better?

Natural collagen is a huge molecule. The body cannot absorb it whole, so makers cut it into small pieces. The Da number on the box tells you how small those pieces are.

1 Da is a unit of weight for molecules

Da is short for dalton. Scientists use it to weigh molecules, which are far too small to see. One dalton is about the weight of one hydrogen atom, the lightest atom there is. A bigger Da number means a bigger piece.

2 Collagen is a chain of beads

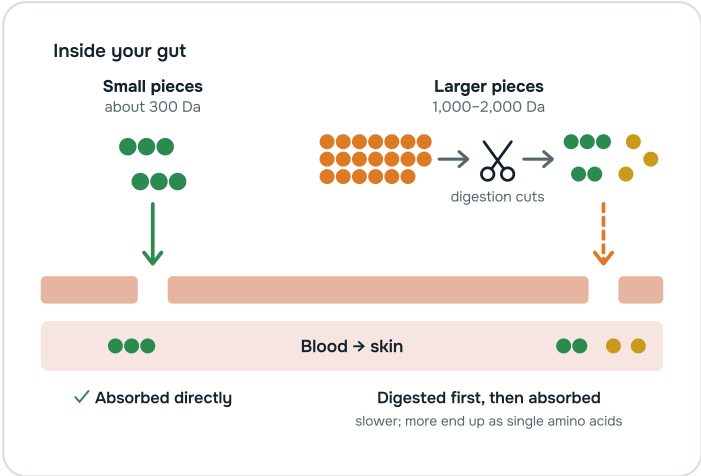
Collagen is built from small building blocks called amino acids. Picture a very long necklace: each bead is one amino acid and weighs about 100 Da. Natural collagen has about 3,000 beads, so it weighs about 300,000 Da.[1] Makers use enzymes to cut this necklace into short pieces.

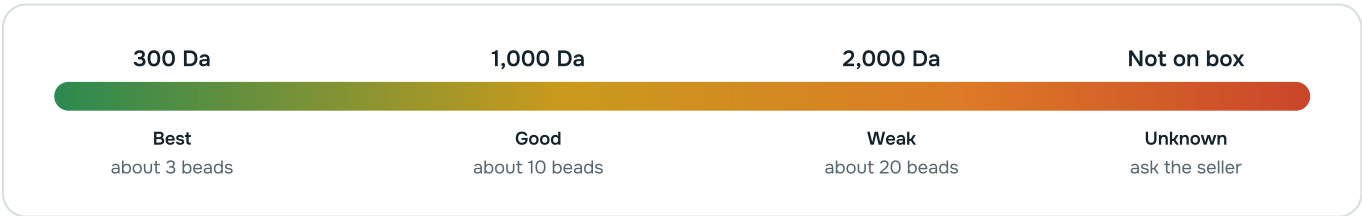
3 Larger pieces are digested before they are absorbed

The gut wall takes up pieces of 2 or 3 beads (about 200–400 Da) with the help of special transporters.[5] Larger pieces are not wasted: digestion cuts them smaller first, and then they are absorbed too. But this takes longer, and more of the collagen ends up as single amino acids instead of the small collagen pieces found in blood in studies.[3]

4 The number on the box is an average

A box that says 300 Da means the pieces weigh 300 Da on average: most are tiny, a few are bigger. A lower average means more of the collagen is already small enough to absorb.



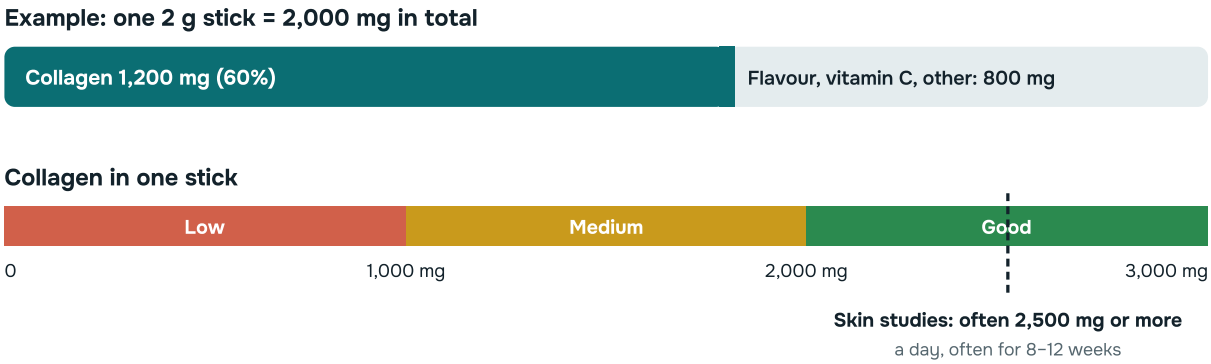


REMEMBER

Look for 평균분자량 (average molecular weight) on the box. 300 Da or lower is the best you can buy.

How much collagen is in one stick?

Small pieces help the body absorb collagen. But the amount matters just as much: a stick with tiny pieces and very little collagen cannot do much.



1 mg and g

mg means milligram. 1,000 mg = 1 g. Stick weight is usually shown in grams, and ingredients in milligrams.

2 Not the whole stick is collagen

A 2 g stick weighs 2,000 mg, but part of it is flavour, vitamin C and other ingredients. Only the collagen part counts. That is why two sticks of the same weight can hold very different amounts of collagen.

3 How to find the amount on the box

Look for the collagen amount per stick (1포당). If the box shows only a percentage, multiply: stick weight in mg × collagen % = collagen in mg. Example: 2,000 mg × 60% = 1,200 mg. Some boxes print the total for the whole box, so check that the number is for one stick.

4 How much is enough

In skin studies, people often took 2,500 mg or more of collagen a day.[67] On our scale, under 1,000 mg per stick is low, 1,000–2,000 mg is medium and 2,000 mg or more is good.

REMEMBER

Choose at least 1,000 mg of collagen per stick. 2,000 mg or more is better.

How the comparison sheet scores products

Each product gets four scores out of 10, calculated from the amount in one stick. The overall score is their weighted average.

Collagen per stick: mg ÷ 250 (2,500 mg or more = 10)	**2 parts (29%)**
Piece size: 300 Da or less = 10, up to 600 = 8, up to 1,000 = 6, up to 2,000 = 4, not shown = 3	**2 parts (29%)**
Vitamin C: mg ÷ 10 (100 mg or more = 10)	**1.5 parts (21%)**
Clean formula: our judgement of whether extras are present in useful amounts and whether the stick is mostly filler	**1.5 parts (21%)**

Data come from the product labels. Where the label shows only a percentage, we multiplied it by the stick weight. MIPPEUM's figures and HURUM Small Molecular's 300 Da come from seller listings.

Limits: we did not test the products in a laboratory. The thresholds are our own choices, guided by the studies cited in this guide. A higher score means a stronger label, not a proven effect on skin.

Why vitamin C matters for collagen

Vitamin C does not turn into collagen. But the body cannot build collagen without it. That is why good collagen sticks include it.



1 Skin cells build collagen

Special skin cells make new collagen fibres. They use amino acids from food and from collagen supplements as building blocks, like bricks.

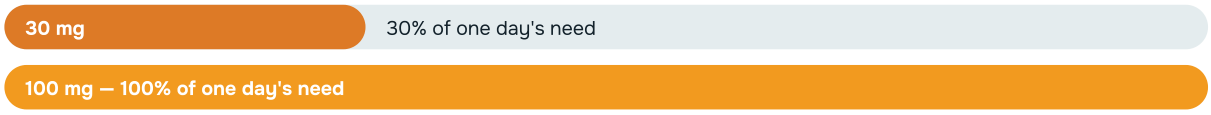
2 Vitamin C is the builder's tool

To join the chains into a strong, stable fibre, these cells use two enzymes, and both need vitamin C to work.[8] Without it, the new collagen is weak and falls apart. This is why people with a severe lack of vitamin C (scurvy) get bleeding gums and wounds that do not heal.

3 The body cannot store it

The human body cannot make vitamin C and stores very little of it. We need a fresh supply every day from food or supplements.

How much of one day's vitamin C need does a stick cover?



Daily reference value: Korea 100 mg, USA 90 mg, Europe 80 mg.

4 How much to look for

Korea sets the daily reference value for vitamin C at 100 mg.[9] A stick with about 100 mg covers the full day. A stick with 30 mg covers less than a third. If vitamin C is not listed on the box, assume there is none.

5 What vitamin C does not do

Vitamin C does not help collagen get into the blood. Its job starts later, inside the skin, where it helps build new collagen. More than 100 mg is fine, but it does not raise the score in our ranking.

REMEMBER

Look for about 100 mg of vitamin C in each stick.

EXTRA INGREDIENTS

Extra ingredients: is there enough to matter?

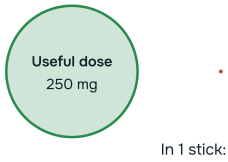
Many boxes advertise extra ingredients in big letters, such as glutathione, biotin or hyaluronic acid. An ingredient can only help if there is enough of it. Below, the green circle shows the amount used in a study, and the red dot shows how much is in one stick.



Glutathione

VITASUB

An antioxidant that boxes often link to "whitening". In the best-known trial, adults took 250 or 1,000 mg a day for 6 months to raise glutathione levels in the body.[12]



MATH COMPARISON

1,250×

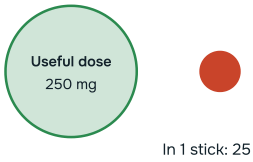
The amount used in the study is 1,250 times what one stick contains. This compares amounts only; it is not a suggestion to take more sticks.



Glutathione

BOTO

An antioxidant that boxes often link to "whitening". In the best-known trial, adults took 250 or 1,000 mg a day for 6 months to raise glutathione levels in the body.[12]



MATH COMPARISON

10×

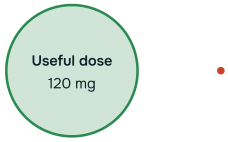
The amount used in the study is 10 times what one stick contains. This compares amounts only; it is not a suggestion to take more sticks.



Hyaluronic acid

HURUM Small Molecular

A substance that holds water in the skin. In the study on dry skin, people took 120 mg a day.[13]



MATH COMPARISON

300×

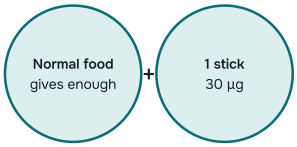
The amount used in the study is 300 times what one stick contains. This compares amounts only; it is not a suggestion to take more sticks.



Biotin

BB LAB Biotin, HURUM Biotin

A B vitamin (vitamin B7). An adult needs about 30 µg a day.[14] µg means microgram: 1 mg = 1,000 µg.



No extra effect

One stick gives a full day of biotin, but normal food already gives enough. There is no good evidence that extra biotin improves skin or hair in people who are not lacking it.[15]

REMEMBER

A famous ingredient name is not a sign of quality. Check the amount in mg and compare it with the amount used in studies.

CHECKLIST

6 questions to ask before you buy

Use these questions for any collagen product, including ones not in this guide.

Is it fish collagen?

Fish collagen is mostly Type I, the type found in skin. It contains no pork or beef.

Is the Da printed on the box, and is it 300 Da or lower?

Smaller pieces are absorbed more easily. If the box does not show Da, ask the seller.

Is there at least 1,000 mg of collagen in one stick?

2,000 mg or more is better. Skin studies often used 2,500 mg or more a day.

Is there about 100 mg of vitamin C in one stick?

The body needs vitamin C to build collagen. 100 mg covers one full day.

Are the extra ingredients there in useful amounts?

Check the mg. A famous name in a tiny amount does nothing.

Are the amounts shown for one stick?

Some boxes show the total for the whole box. Always compare one stick with one stick.

Sources

1. Shoulders MD, Raines RT. Collagen structure and stability. *Annu Rev Biochem.* 2009;78:929–958.
2. Varani J, Dame MK, Rittie L, et al. Decreased collagen production in chronologically aged skin. *Am J Pathol.* 2006;168(6):1861–1868.
3. Iwai K, Hasegawa T, Taguchi Y, et al. Identification of food-derived collagen peptides in human blood after oral ingestion of gelatin hydrolysates. *J Agric Food Chem.* 2005;53(16):6531–6536.
4. Shigemura Y, Iwai K, Morimatsu F, et al. Effect of prolyl-hydroxyproline (Pro-Hyp), a food-derived collagen peptide in human blood, on growth of fibroblasts from mouse skin. *J Agric Food Chem.* 2009;57(2):444–449.
5. Daniel H. Molecular and integrative physiology of intestinal peptide transport. *Annu Rev Physiol.* 2004;66:361–384.
6. Proksch E, Segger D, Degwert J, et al. Oral supplementation of specific collagen peptides has beneficial effects on human skin physiology: a double-blind, placebo-controlled study. *Skin Pharmacol Physiol.* 2014;27(1):47–55.
7. Pu SY, Huang YL, Pu CM, et al. Effects of oral collagen for skin anti-aging: a systematic review and meta-analysis. *Nutrients.* 2023;15(9):2080.
8. Pullar JM, Carr AC, Vissers MCM. The roles of vitamin C in skin health. *Nutrients.* 2017;9(8):866.
9. Ministry of Food and Drug Safety (Korea). Standards for Labeling of Foods (식품등의 표시기준): nutrient reference values.
10. U.S. Food and Drug Administration. Daily Value on the Nutrition and Supplement Facts Labels.
11. Regulation (EU) No 1169/2011 on the provision of food information to consumers, Annex XIII: nutrient reference values.
12. Richie JP Jr, Nichenametla S, Neidig W, et al. Randomized controlled trial of oral glutathione supplementation on body stores of glutathione. *Eur J Nutr.* 2015;54(2):251–263.
13. Kawada C, Yoshida T, Yoshida H, et al. Ingested hyaluronan moisturizes dry skin. *Nutr J.* 2014;13:70.
14. Ministry of Health and Welfare, The Korean Nutrition Society. 2020 Dietary Reference Intakes for Koreans.
15. Patel DP, Swink SM, Castelo-Soccio L. A review of the use of biotin for hair loss. *Skin Appendage Disord.* 2017;3(3):166–169.

Scores are based on the amounts printed on each product's label or seller listing. This guide compares product labels to help with buying decisions. It is not medical advice.